Mystery Meat Analysis

BIOLM0051

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Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

Introduction

Increased global trade has increased the prevalance of

Methods

Pipeline Overview

Results

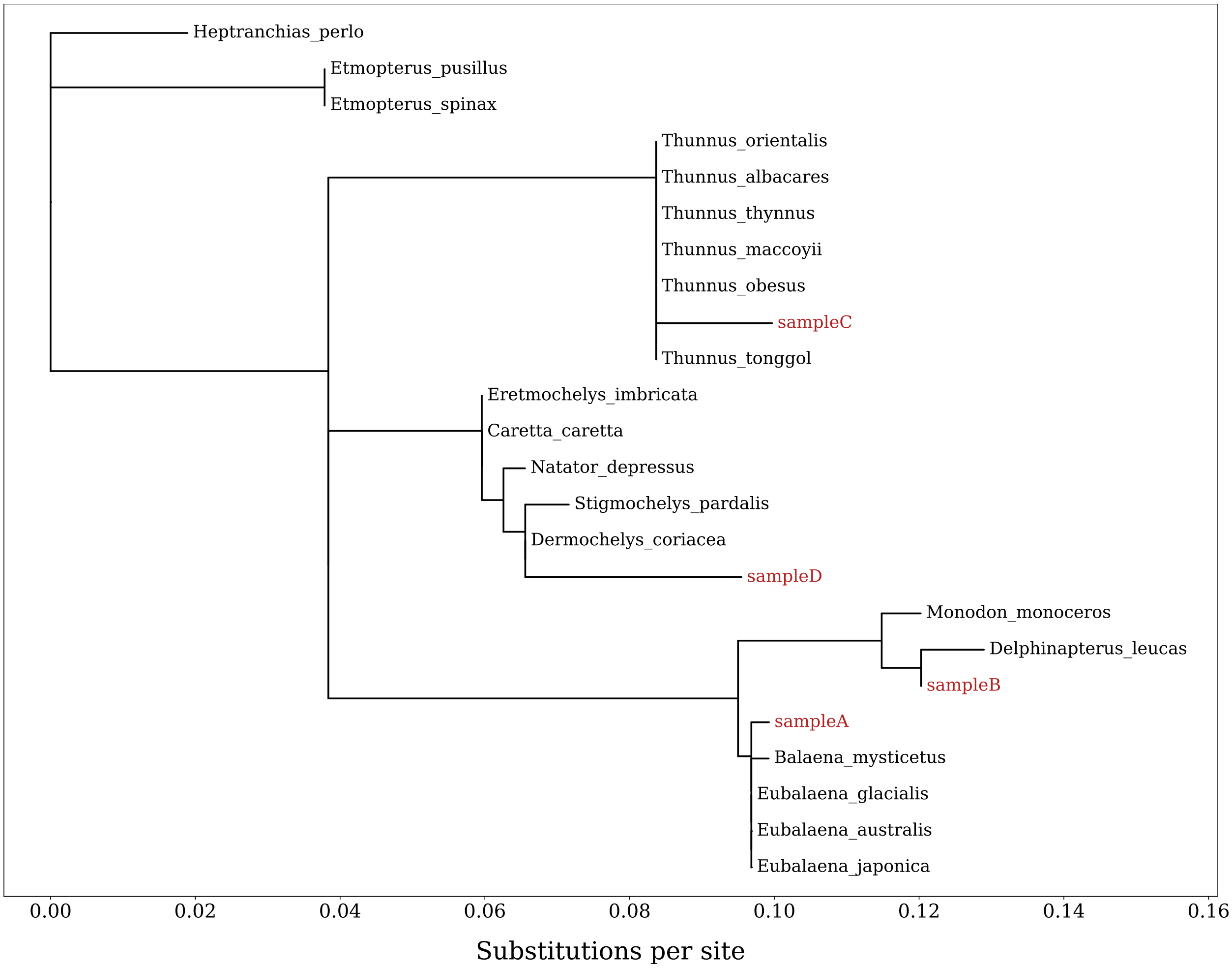

Figure 1: Maximum-likelihood phylogenetic tree with placement of unknown meat samples and their top BLAST hits

Discussion

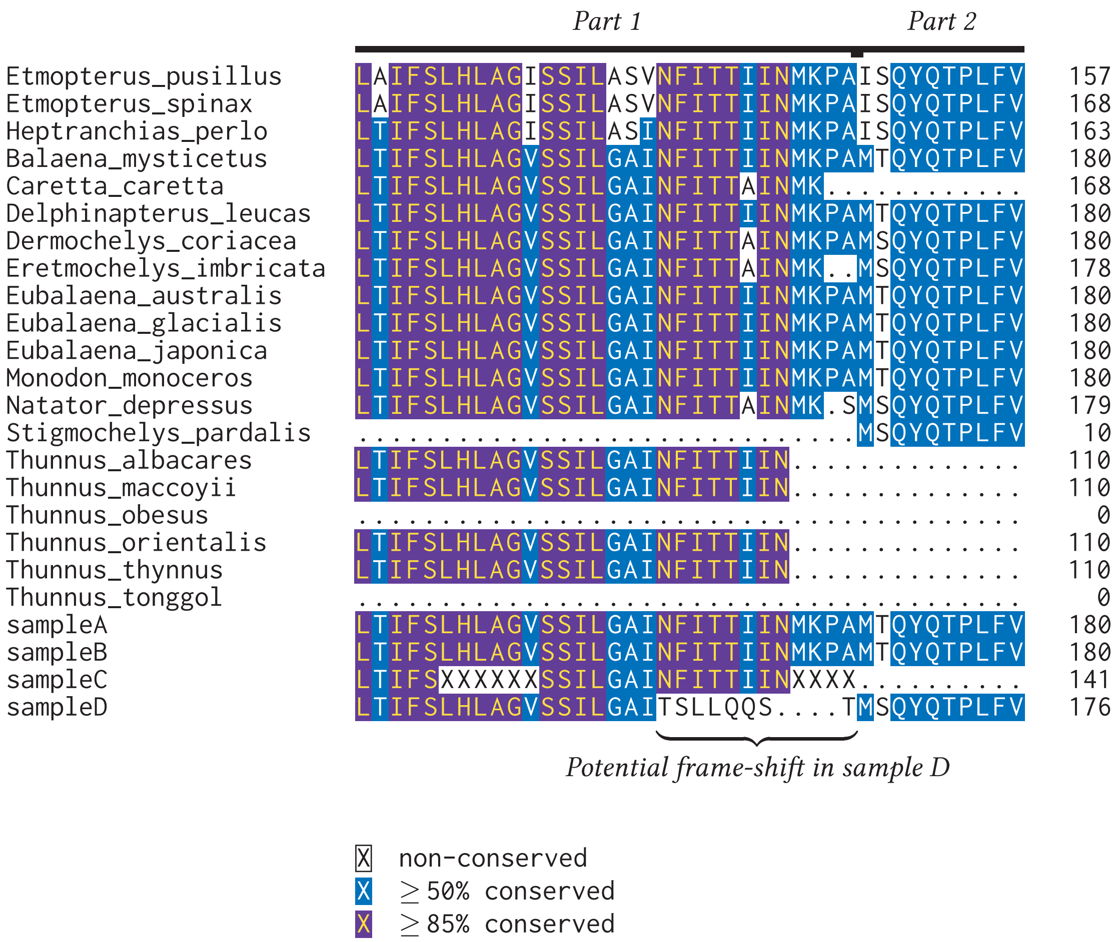

Figure 2: Window of supermatrix alignment with potential frame-shift error in `sampleD` near the highly-conserved `NFITTIIN` region.

```
Dermochelys_c     441  CACCTAGCTGGTGTTTCATCAATTTTAGGAGCTATTAACTTCATTACTAC     490
                       |||||||||||||||||||||||||||||||||| |||||||||||||
sampleD_part1     451  CACCTAGCTGGTGTTTCATCAATTTTAGGAGCTATT-ACTTCATTACTAC     499
```

↑

Figure 3: Needleman-Wunsch global pairwise alignment at the tail end of `sampleD_part1` against closest reference sequence (*Dermochelys coriacea*). The alignment gap indicates a possible base miss at position 486 in `sampleD_part1` during sequencing.

References

Supplementary Data

Supplementary data and all scripts can be found within the GitHub repository for this project: https://github.com/archiebenn/BIOLM0051_analysis.git

Appendix: Exploratory Figures for Manual Correction of Suspected Sequencing Error

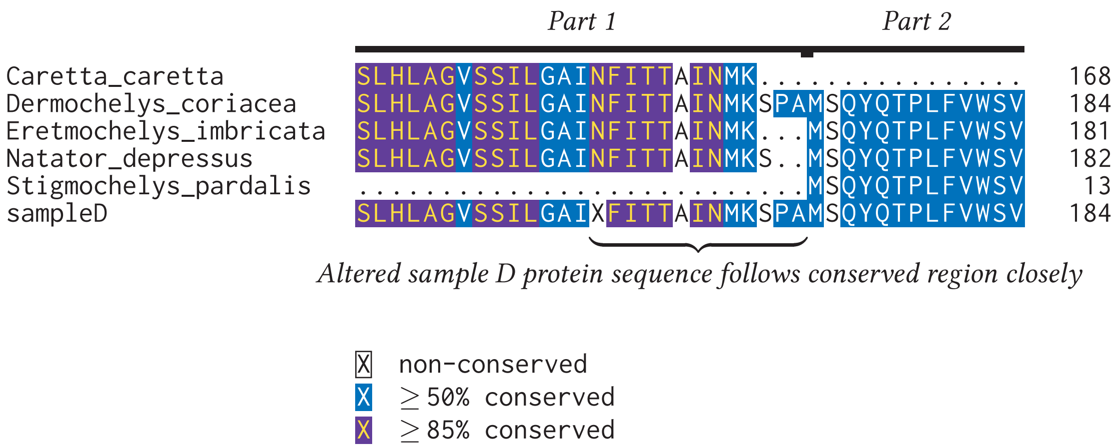

Figure 4: Subset of supermatrix alignment with `sampleD` and innner clade after inserting one ambiguous base (N) to the nucleotide sequence of `sampleD_part1`. This base resolves the apparent frame-shift, resulting in a protein sequence which aligns much more closely to highly-conserved `NFITTIIN` region compared to in Figure 2.

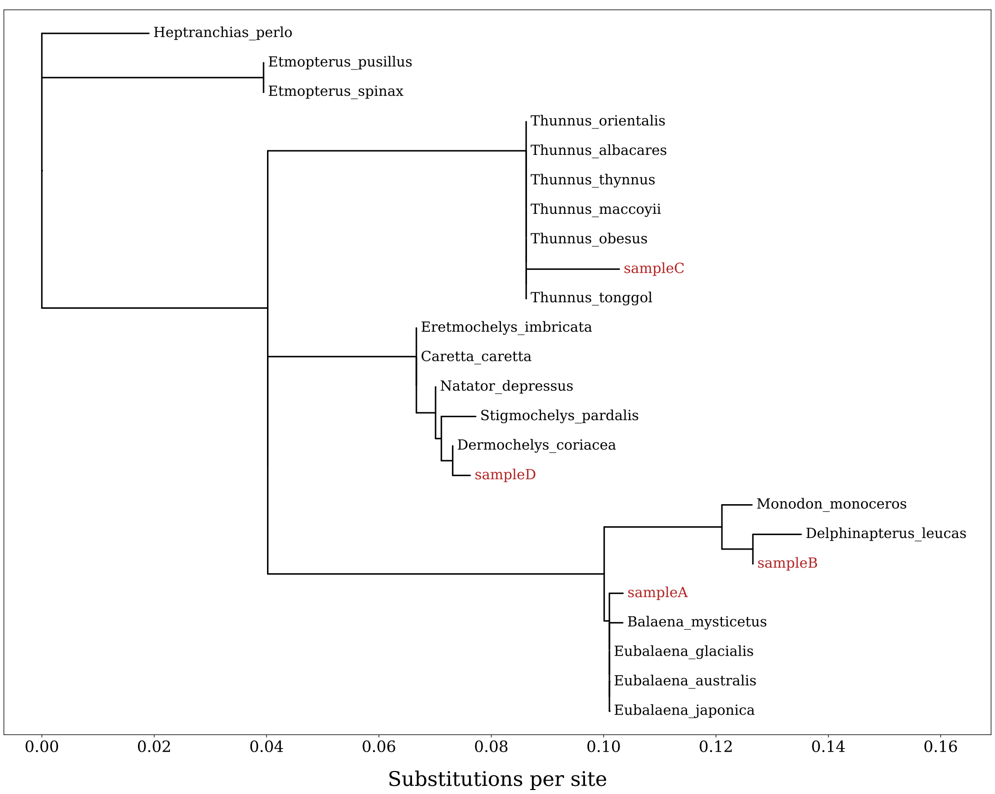

Figure 5: Maximum-likelihood phylogenetic tree after manually corrected alignment in which one ambiguous base (N) was inserted into `sampleD_part1`. This tree is shown to demonstrate the impact of sequencing error and downstream frame-shift on branch lengths compared to Figure 1.